Nicolas Santini

Cloud & Data Engineer | Google Cloud

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PROFESSIONAL SUMMARY

Cloud and Data Engineer working as an independent contractor on Google Cloud, with close to three years of production experience designing, building and operating ETL/ELT and event-driven data pipelines in Python and SQL. Most recently the sole engineer on the migration of a legacy MDM platform to Google Cloud for a large Italian multi-brand retail group (8 store brands, 15M+ customer records), where every transformation rule was reconstructed from an undocumented system and proven correct field by field before cutover. Earlier, high-volume API ingestion at hundreds of thousands of records per cycle and a MongoDB to BigQuery migration that cut query time by about 90%. Alongside engineering, I design and deliver hands-on generative AI training for companies and schools. Google Cloud Professional Machine Learning Engineer. Italian VAT registered, open to remote or hybrid work. Italian native, English C1.

TECHNICAL SKILLS

Google Cloud: BigQuery, Dataflow (Apache Beam), Pub/Sub, Cloud Run, Cloud Functions, Cloud Workflows, Cloud Storage, Cloud SQL (PostgreSQL), Secret Manager, Cloud Build, Cloud Monitoring and Logging

Pipelines & orchestration: ETL/ELT design, event-driven and serverless architecture, Apache Airflow, data validation and reconciliation

Languages: Python (Pandas, NumPy, FastAPI, Django), SQL

Databases: BigQuery, PostgreSQL, MySQL, MongoDB, Firestore, DynamoDB

AWS: Lambda, S3, DynamoDB, CloudWatch (about two years)

Engineering practice: REST APIs with OAuth2, Docker, Git, Bitbucket, Git-based CI/CD

Generative AI: Google Gemini, NotebookLM, prompt design, AI enablement for non-technical teams

WORK EXPERIENCE

AI Trainer for Teachers | MakerDojo

Oct 2026 – Present | Freelance

Freelance engagement run alongside engineering work, for an education organisation bringing coding, robotics and AI into schools.

- Designed from scratch, at the client's request, a hands-on generative AI curriculum (Google Gemini, NotebookLM) for teachers of a public school network in Piacenza, then reworked it around the school's constraints after the first review.
- Tailored the curriculum to four subject areas and three school levels, from preschool to middle school, within a 38-session programme for 19 teacher groups.
- Deliver four-hour in-person workshops where teachers build and verify real classroom materials; produced facilitator guides, slide decks and a five-part prompting framework, with data-protection rules for minors built into every session.

Cloud & Data Engineer | JAKALA

Mar 2026 – Sep 2026 | Contract

Sole technical owner of the migration of a legacy Informatica PowerCenter MDM platform to a cloud-native architecture on Google Cloud, for a large Italian multi-brand retail group: 8 store brands, over 15 million customer records.

- Redesigned the data layer around event-driven serverless pipelines on Cloud Run, Cloud Functions and Cloud Workflows, replacing scheduled batch ETL with near real-time processing that reacts to source data changes.
- Reverse engineered undocumented legacy logic (multi-source record matching, survivorship rules, golden record definitions), validated it with business owners and rebuilt it as modular, testable pipeline components.
- Ran the new platform in parallel with the legacy one and built a reconciliation layer comparing both outputs record by record and field by field; cutover was gated on every difference being classified as a legacy defect or a regression.
- Implemented data validation, schema monitoring and operational alerting so that quality issues surfaced before reaching the business.
- Built the target persistence layer on Cloud SQL (PostgreSQL), with credentials, security policies and automatic rotation managed through Secret Manager.
- Treated marketing consent data as a compliance concern, not only a quality one, ensuring each consent stayed bound to the correct identity through matching and merge.
- Defined a cutover strategy that moved downstream enterprise applications off the legacy platform with no service interruption, keeping the legacy system as a rollback path.

Innovation Engineer | WEBRANKING Apr 2024 – Mar 2026

Data engineering across 50+ client analytics accounts, from high-volume ingestion pipelines to the redesign of the core analytics stack.

High-volume, event-driven ingestion

- Built and operated a continuous pipeline ingesting search keyword intelligence from an external provider API, handling hundreds of thousands of keywords per cycle.
- Coordinated long-running processing services through Pub/Sub, which drove orchestration and downstream notification of completed batches, with results landed in BigQuery.
- Handled third-party ingestion at volume: provider rate limits, partial failures, retries and inconsistent schemas across sources.

Analytics platform and pipelines

- Led the migration of the core analytics stack from MongoDB to BigQuery, cutting query time by about 90% and making multi-year analyses, previously impractical, part of the routine.
- Architected and operated production pipelines with Python, Dataflow and Airflow, turning raw marketing signals into clean datasets used daily by analytics teams.
- Designed and shipped REST APIs in FastAPI with OAuth2, automating data integration between internal platforms, Google Analytics and Meta Business Manager.
- Built and operated serverless components on AWS (Lambda, S3, DynamoDB) with CloudWatch monitoring and alerting.

Engineering practice and enablement

- Established Git-based CI/CD release workflows with Cloud Build across engineering teams and containerised workloads with Docker to standardise execution environments.
- Designed and delivered a full-day, on-site AI training programme for about 20 senior staff at Tecnomat, on AI in business processes and task automation, closing with a hands-on workshop.
- Acted as the technical bridge between marketing strategists and engineering, scoping requirements and assessing feasibility before development.

CERTIFICATIONS

- **Google Cloud Professional Machine Learning Engineer**, January 2026, verifiable on Credly.
- **Google Cloud Professional Data Engineer**, in preparation, target late 2026.

EDUCATION

- **BSc in Computer Engineering**, 106/110, University of Modena and Reggio Emilia, Oct 2020 – Feb 2024. Thesis: comparative study of relational and non-relational database models, the same question later answered in production moving an analytics workload from MongoDB to BigQuery.
- **Erasmus+ exchange in Computer Science**, Kielce University of Technology, Poland, Sep 2022 – Feb 2023.

SELECTED PERSONAL PROJECTS

- **Serverless market digest pipeline.** Event-driven pipeline on Pub/Sub and Cloud Functions that fetches market data on a schedule, computes daily moving statistics across a watchlist and delivers a formatted summary.
- **Multi-source training assistant.** Aggregates training, sleep and recovery data from several fitness providers into one normalised model and writes a periodised plan to Google Calendar, handling OAuth2 token refresh, per-provider rate limits and inconsistent schemas.
- **Vision-driven data extraction pipeline.** Turns unstructured screenshots into structured records with an LLM vision model, with a confidence threshold and a manual review path for ambiguous extractions.

VOLUNTEERING

- **Causes Coordinator**, Erasmus Student Network Modena e Reggio Emilia, Sep 2025 – Present. Runs the causes programme for incoming international students (health and wellbeing, social inclusion, sustainability), from mental health workshops to sports events and their safety protocols.
- **Volunteer and Event Manager**, ESN Modena e Reggio Emilia, Feb 2023 – Sep 2025. Ran events end to end and authored the section's official activity reports in English.
- **Volunteer**, de Amicitia and Legambiente, Colunga, Spain, Aug 2023. Three weeks on Huerta y Tech III, bringing technology into small-scale agriculture.

LANGUAGES & BEYOND WORK

Languages: Italian (native), English (C1, professional working proficiency), French and Spanish (A2).

Beyond work: gym, endurance sport, amateur football and volleyball. Training in blocks toward two 2027 goals: the 100 km del Passatore road ultra in spring and a full Ironman in autumn.